

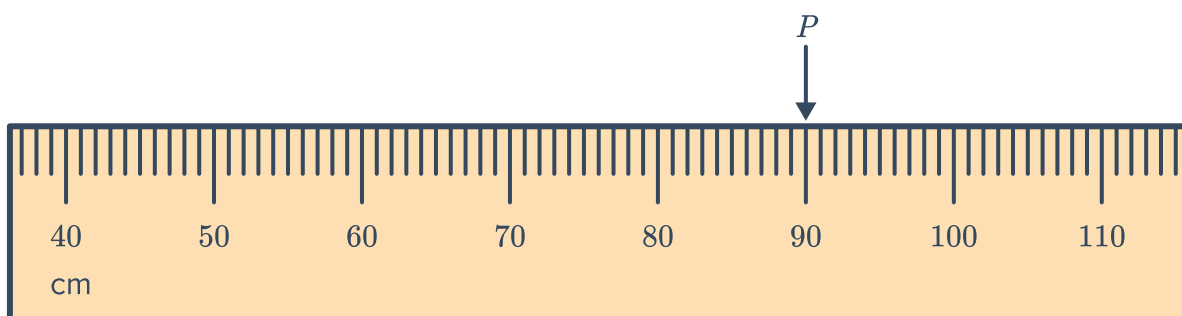
Absolute error



- 1 State the absolute error of the following:
 - a A measurement of 30 kg, which was measured to the nearest kilogram.
 - b A measurement of 400 g, which was measured to the nearest gram.
 - c The amount \$700, which was calculated to the nearest \$10.
 - 2 A chef measures the amount of olive oil to be used in a cup marked in millilitres. Find the absolute error of any measurement she makes.
 - 3 State the absolute error of each of the following measurements obtained from measuring devices:
 - a 18.9 mL
 - b 269 mm
 - c 9.9 g
 - d 18 s
 - e 15.12 kg
 - f 17.8 m
 - g 117.38°C
 - 4 The number of people at a concert is approximately 5050. Explain why we cannot find the absolute error of this approximation.
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The limits of accuracy

- 5 Consider the tape measure below:

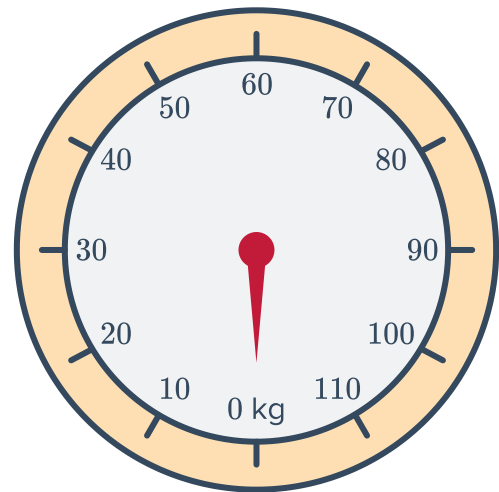


- a What is the smallest unit marked on the tape?
- b What should be recorded for measurement P ?
- c What is the absolute error of measurement P ?
- d State whether the following could be the actual measurement of P :
 - i 89.02 cm
 - ii 90.64 cm
 - iii 89 cm
 - iv 90.03 cm

6 Consider the following scale:



- a What is the smallest unit labelled on the scale?
- b What is the absolute error?
- c If a certain object is measured at 70 kg, what are the lower and upper bounds of this measurement?



7 Consider the speedometer below:

- a At what speed is being indicated on the speedometer?
- b What is the absolute error of this speed reading?
- c What would the maximum speed limit need to be to ensure the driver is not speeding?



8 For the following measurements, find:

- i The upper bound
 - ii The lower bound
- a A distance measured to be 13.45 km.
 - b A distance measured to be 6.4 km.
 - c A height measured to be 5 m.
 - d A bag of sugar weighs 14 kg to the nearest 10 grams.
 - e Puncak Jaya, Indonesia's highest mountain, is 4884 m high rounded to the nearest metre.
 - f The cost of a CD lie if it is known to be \$50 correct to the nearest \$5.

9 The length of a piece of rope is measured to be 19.99 m using a ruler. What is the upper bound of the largest possible length of this rope?

- 10 The height of a tower is measured as 2.1 m. What is the shortest possible height of the tower?
- 11 State the maximum and minimum possible number of the following:
- a A town's population is estimated to be 170 people, to the nearest 10 people.
 - b A stack of paper is estimated to have 3000 sheets to the nearest 100 sheets.
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Further calculations (Extended)

- 12 $c = 4$ and $d = 8$ have both been rounded to the nearest whole number.
- a Calculate the lower bound of the value of $c \times d$.
 - b Calculate the upper bound of the value of $\frac{c}{d}$.
- 13 A school field has a 100 m track that is measured to the nearest metre. The record for the fastest 100 m sprint is 13.9 seconds rounded to the nearest 0.1 seconds.
- a What is the fastest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.
 - b What is the slowest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.
- 14 A circle has a radius of 7 cm, correct to the nearest centimetre.
- a Calculate its area correct to one decimal place.
 - b What are the smallest and largest possible areas? Round your answer to one decimal place.
 - c What is the maximum possible absolute error in the calculation?
 - d What is the maximum possible percentage error in the calculation? Round your answer to one decimal place.
- 15 A field has dimensions 15.4 m $\times$ 17.6 m, to the nearest 10 cm.
- a What are the upper and lower bounds of the area of the field?
 - b What is the upper and lower bounds of the perimeter of the field?

16 A cube has a side length of 10 cm rounded to the nearest centimetre.



- a Find the lower and upper bounds of the side length.
- b Find the area of one face of the cube, using the given side length.
- c Find the lower and upper bounds of the area of one face.
- d Find the volume of the cube using the given side length.
- e Find the lower and upper bounds of the volume of the cube.